Requirement Specification for a Linear Equation Calculator

Muhammad Ahmed Qazi

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Abstract

This article presents an initial draft of the software requirements for a calculator application capable of performing arithmetic operations and solving basic linear equations.

1 Introduction

Conventional pocket calculators evaluate arithmetic expressions but cannot resolve an unknown quantity presented in equation form. The proposed calculator extends standard arithmetic functionality with a symbolic solver for basic linear equations, allowing a user to enter an equation directly and obtain its solution.

1.1 User Requirements

Following are the requirements that our system fulfils:

1. **Arithmetic:** A user must be able to evaluate expressions involving addition, subtraction, multiplication and division.
2. **Equation entry:** A user must be able to enter an equation containing an equality sign and one or more unknowns.
3. **Solving:** The system must determine the value of each unknown and report the following:
    - A unique solution, where one exists.
    - That the equation holds for every value of the unknown.

- That the equation has no solution at all.

4. **Error handling:** A user must be informed when the input is malformed or when division by zero occurs.
5. **History:** A user should be able to view previously solved equations.

2 Supported Equations

The solver is able to handle the classes of equations shown below.

- **Single unknown on one side:**

$$2x + 5 = 15 \qquad \Rightarrow \qquad x = 5$$

- **Unknown on both sides:**

$$3x - 4 = x + 6 \qquad \Rightarrow \qquad x = 5$$

- **Parenthesised terms:**

$$2(x + 3) = 10 \qquad \Rightarrow \qquad x = 2$$

- **Fractional coefficients:**

$$\frac{x}{2} + 1 = 4 \qquad \Rightarrow \qquad x = 6$$

- **Simultaneous equations in two unknowns:**

$$x + y = 5, \quad 2x - y = 1 \qquad \Rightarrow \qquad x = 2,\ y = 3$$

Quadratic, trigonometric and transcendental equations lie outside the scope of this draft.

3 Software Attributes

Our calculator application has the following software attributes:

- Functionality: arithmetic evaluation, linear equation solving and history retrieval.

- Reliability: every solution is verified by substitution before it is displayed to the user.
- Usability: equations are entered in ordinary mathematical notation, requiring no specialised syntax.
- Portability: the application is deployed both as a desktop program and as an Android mobile application.

4 Conclusion

This draft establishes the scope of a calculator that treats an equation as a first-class input rather than a sequence of keystrokes. By restricting the solver to linear forms, the system remains small enough to specify precisely while still covering the equations most commonly encountered by its intended users. Subsequent revisions may extend the grammar to higher-degree polynomials once the linear case has been validated.